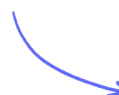


Symmetry & Shapes

Rotational Symmetry / Lines of Symmetry / 2D Shapes / 3D Shapes / Planes of Symmetry

Easy (6 questions)	/8
Medium (4 questions)	/7
Hard (3 questions)	/3
Total Marks	/18

Scan here to return to the course
or visit [savemyexams.com](https://www.savemyexams.com)



Easy Questions

1 A quadrilateral has

- exactly 2 lines of symmetry

and

- rotational symmetry of order 2.

Write down the mathematical name of the quadrilateral.

(1 mark)

2 Write down the number of lines of symmetry of a kite.

(1 mark)

3 (a) Complete each statement.

A quadrilateral with only one pair of parallel sides is called a

(1 mark)

(b) An angle greater than 90° but less than 180° is called

(1 mark)

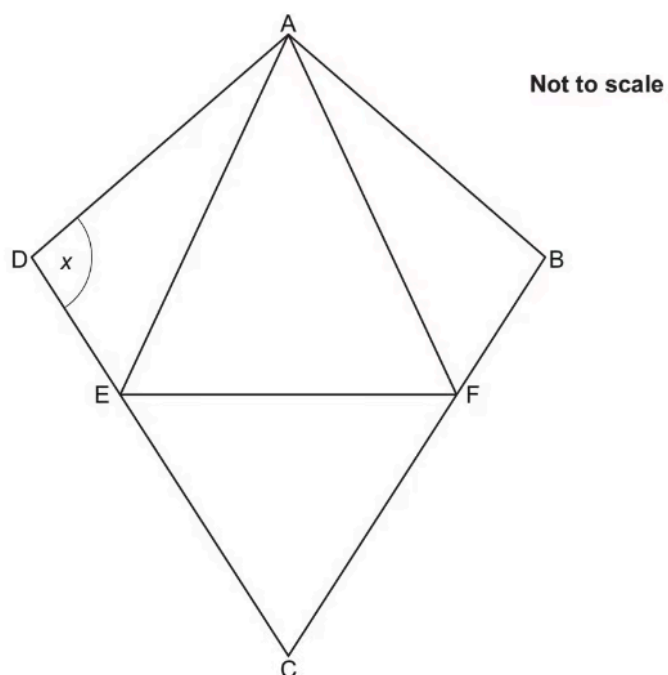
4 Circle the solid that has 7 vertices.

hexagonal prism	hexagon-based pyramid	pentagonal prism	pentagon-based pyramid
--------------------	--------------------------	---------------------	---------------------------

(1 mark)

5 The diagram shows a kite, ABCD.

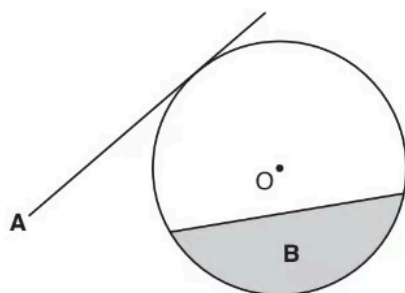
AFE and CEF are equilateral triangles.



Write down a mathematical name for quadrilateral AFCE.

(1 mark)

6 The diagram shows a circle, centre O.



Write down the mathematical name of

i) line **A**,

[1]

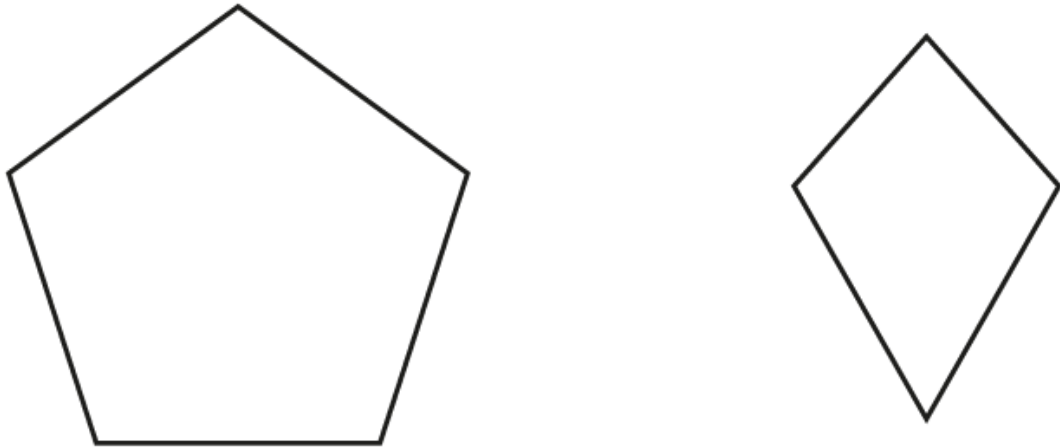
ii) shaded region **B**.

[1]

(2 marks)

Medium Questions

1 (a)



The diagram shows a regular pentagon and a kite.
Complete the following statements.

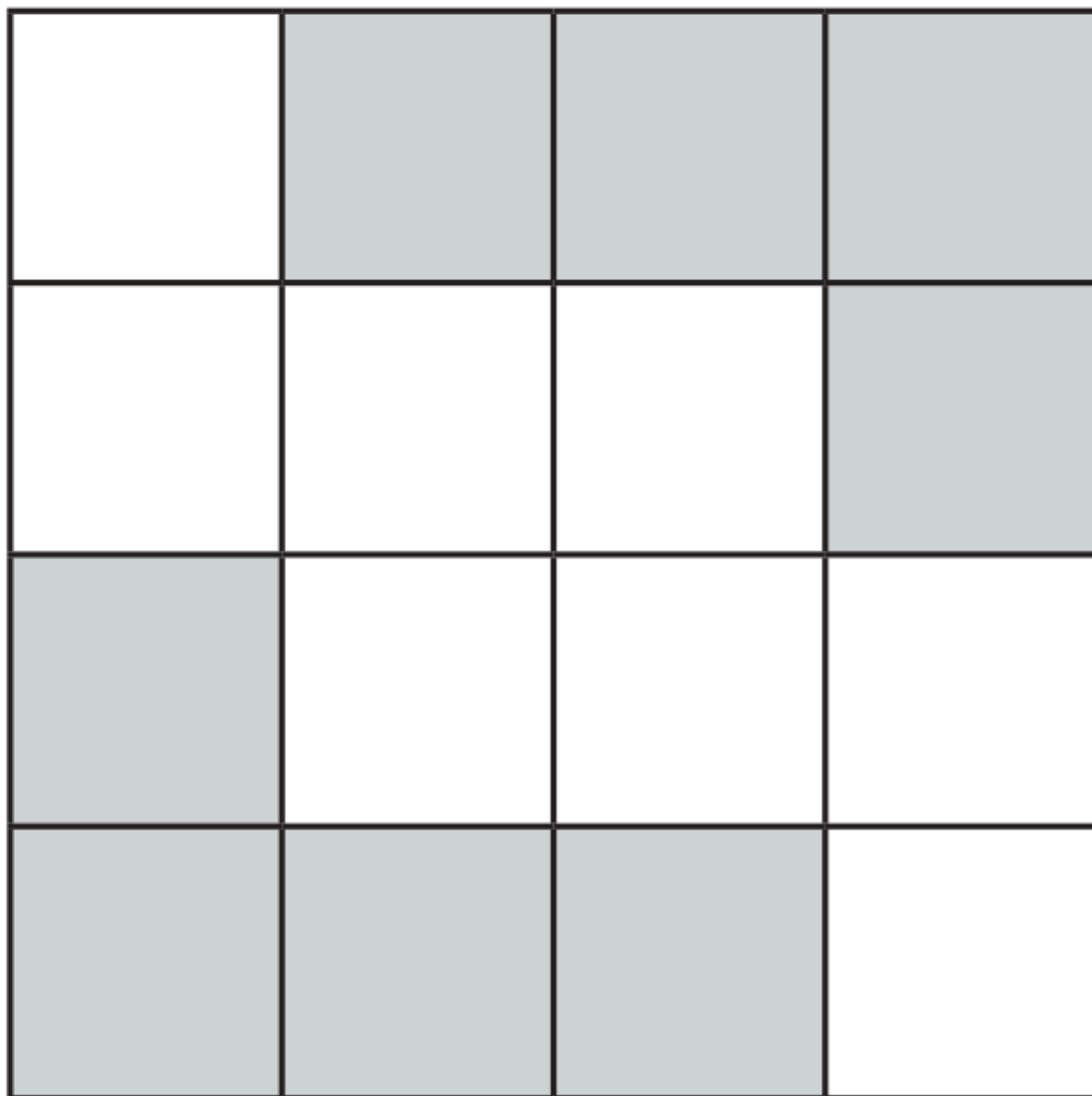
The regular pentagon has lines of symmetry.

(1 mark)

(b) The kite has rotational symmetry of order

(1 mark)

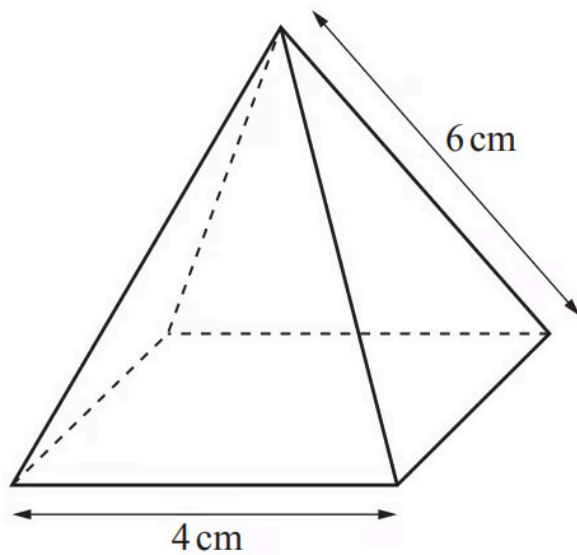
2



Write down the order of rotational symmetry of the diagram.

(1 mark)

3



NOT TO
SCALE

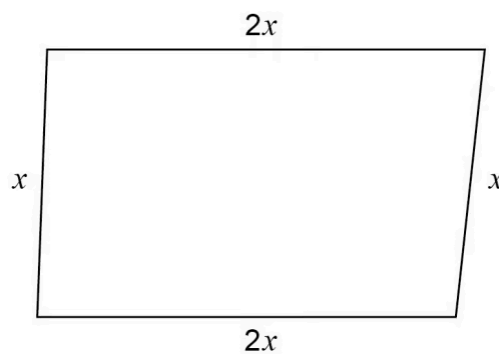
The diagram shows a pyramid with a square base.
The triangular faces are congruent isosceles triangles.

Write down the number of planes of symmetry of this pyramid.

(1 mark)

4 Here is a **sketch** of a quadrilateral.

All lengths are in centimetres.



Not drawn
accurately

Tick **one** box for each statement.

	True	May be true	Not true
The quadrilateral is a rectangle			
The quadrilateral is a parallelogram			
The quadrilateral is a rhombus			
The quadrilateral is a kite			

(3 marks)

Hard Questions

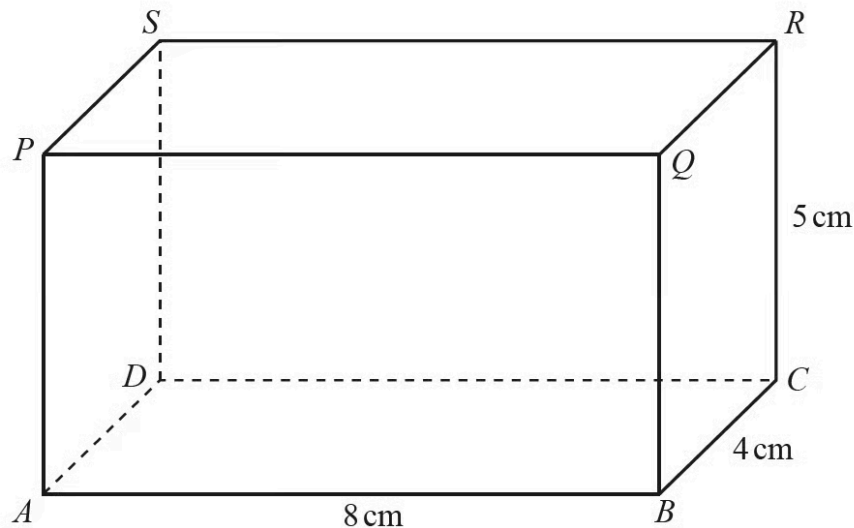
- 1 The cross section of a prism has n sides.

Circle the expression for the number of edges of the prism.

$2n$	$3n$	$n + 2$	$2n + 3$
------	------	---------	----------

(1 mark)

2



NOT TO
SCALE

The diagram shows a cuboid.

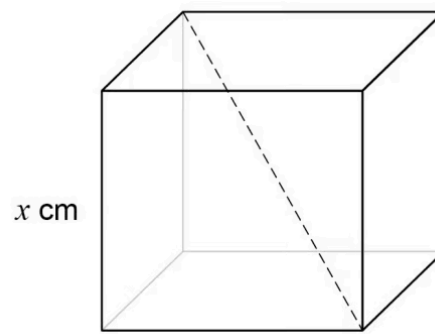
$AB = 8$ cm, $BC = 4$ cm and $CR = 5$ cm.

Write down the number of planes of symmetry of this cuboid.

(1 mark)

- 3 Here is a cube with edge length x cm

One diagonal is shown.



The total length, in centimetres, of the edges of the cube is a multiple of 18

Circle the correct statement.

x is a whole number	x is not a whole number	x might be a whole number
-----------------------	---------------------------	-----------------------------

(1 mark)